

Senario ➖

Our SIEM alerted us to a suspicious logon event which needs to be looked at immediately . The alert details were that the IP Address and the Source Workstation name were a mismatch .You are provided a network capture and event logs from the surrounding time around the incident timeframe. Correlate the given evidence and report back to your SOC Manager.

Analysis ➖

We Got Two Files one is Windows Event Logs and the second is Capture of the Traffic during the incident . First i will parse the EventLogs into readable format cause i cant runt it in my Kali Machine

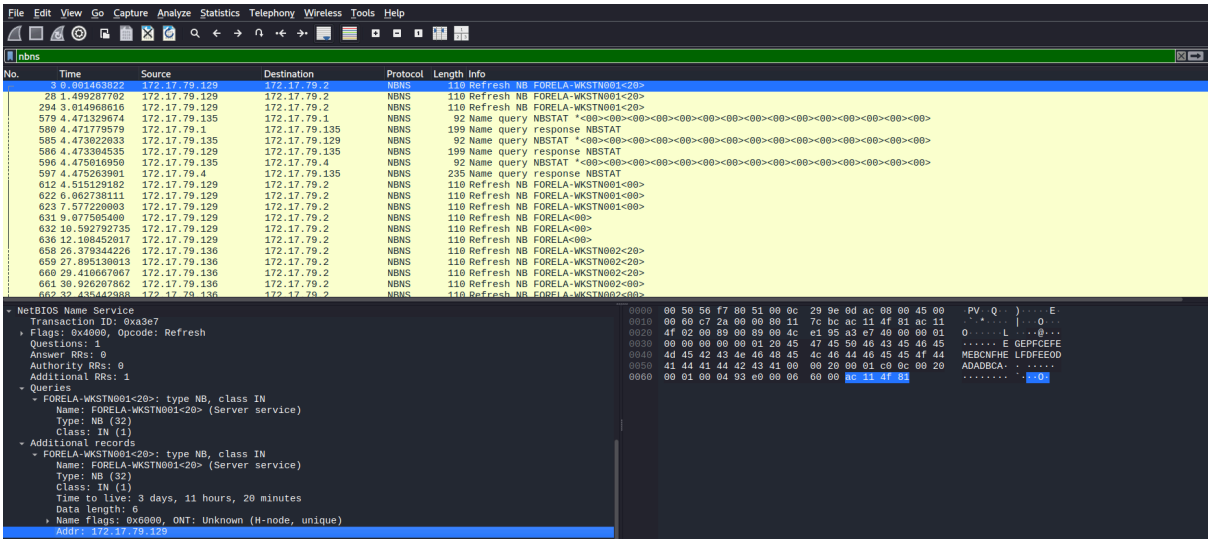
```
┌──(kali㉿kali)-[~/Htb/Reaper/Reaper]
└─$ evtxexport Security.evtx > EventLogs.xml
```

I will open up Wireshark and the event logs in different Terminals and start The Analysis Process . First the important part is the team detected workstation name mismatch with the IP meaning there are some Rules that detected this anomaly so i will go for the EventID 4624 (Successful Logon) so i can see all the users logged in and their associated IP Addresses and only one stands out

```
┌──(kali㉿kali)-[~/Htb/Reaper/Reaper]
└─$ cat EventLogs.xml | grep '(4624)' -A 28 -B 9
```

```
Event number            : 4
Creation time           : Jul 31, 2024 04:55:16.240589700 UTC
Written time            : Jul 31, 2024 04:55:16.240589700 UTC
Event version           : 2
Event level             : Information (0)
Computer name           : Forela-Wkstn001.forela.local
Source name             : Microsoft-Windows-Security-Auditing
Channel name            : Security
Event identifier        : 0x00001210 (4624)
Number of strings       : 27
String: 1               : S-1-0-0
String: 2               : -
String: 3               : -
String: 4               : 0x0000000000000000
String: 5               : S-1-5-21-3239415629-1862073780-2394361899-1601
String: 6               : arthur.kyle
String: 7               : FORELA
String: 8               : 0x000000000064a799
String: 9               : 3
String: 10              : NtLmSsp
String: 11              : NTLM
String: 12              : FORELA-WKSTN002
String: 13              : {00000000-0000-0000-0000-000000000000}
String: 14              : -
String: 15              : NTLM V2
String: 16              : 128
String: 17              : 0x0000000000000000
String: 18              : -
String: 19              : 172.17.79.135
String: 20              : 40252
String: 21              : %%1833
String: 22              : -
String: 23              : -
String: 24              : -
String: 25              : %%1843
String: 26              : 0x0000000000000000
String: 27              : %%1843
```

The Event Number 4 logged a Successful Logon for wkstn001 with the IP Address of 172.17.79.135 we can Search for the Real IP Address for wkstn001 through wireshark traffic and filter on NetBios Name Service (NBNS)



We can see the IP Address of **172.17.79.129** is the legit IP of the machine We can also observe another workstation which is wstn002 with the IP Address of

172.17.79.136 . the attacker Authenticated as **arthur.kyle** according to the EventLogs . that indicates with no doubts that the attacker IP is **172.17.79.135**. Filtering SMB2 traffic in wireshark we can see the Attacker tried to Access the **\\DC01\Trip** share multiple times from the workstation machine as his main Attack machine the previous IP of 172.17.79.135 was the Tool he ran most likely responder that poisoned LLMNR traffic and captured the victim hash then it been relayed into smb connection.

```
1418 112.565901644 172.17.79.136    172.17.79.4      SMB2     152 Tree Connect Request, Tree: '\\DC01\Trip'
1419 112.566148183 172.17.79.4      172.17.79.136    SMB2     130 Tree Connect Response, Error: STATUS_BAD_NETWORK_NAME
1420 112.566315545 172.17.79.136    172.17.79.4      SMB2     152 Tree Connect Request, Tree: '\\DC01\Trip'
1421 112.566424377 172.17.79.4      172.17.79.136    SMB2     130 Tree Connect Response, Error: STATUS_BAD_NETWORK_NAME
1422 112.566608349 172.17.79.136    172.17.79.4      SMB2     152 Tree Connect Request, Tree: '\\DC01\Trip'
1423 112.566717443 172.17.79.4      172.17.79.136    SMB2     130 Tree Connect Response, Error: STATUS_BAD_NETWORK_NAME
1424 112.566899230 172.17.79.136    172.17.79.4      SMB2     152 Tree Connect Request, Tree: '\\DC01\Trip'
1425 112.566984930 172.17.79.4      172.17.79.136    SMB2     130 Tree Connect Response, Error: STATUS_BAD_NETWORK_NAME
1426 112.567344810 172.17.79.136    172.17.79.4      SMB2     152 Tree Connect Request, Tree: '\\DC01\Trip'
1427 112.567420731 172.17.79.4      172.17.79.136    SMB2     130 Tree Connect Response, Error: STATUS_BAD_NETWORK_NAME
1428 112.567600707 172.17.79.136    172.17.79.4      SMB2     152 Tree Connect Request, Tree: '\\DC01\Trip'
1429 112.567678622 172.17.79.4      172.17.79.136    SMB2     130 Tree Connect Response, Error: STATUS_BAD_NETWORK_NAME
1430 112.567878203 172.17.79.136    172.17.79.4      SMB2     152 Tree Connect Request, Tree: '\\DC01\Trip'
1431 112.567986445 172.17.79.4      172.17.79.136    SMB2     130 Tree Connect Response, Error: STATUS_BAD_NETWORK_NAME
1432 112.568175286 172.17.79.136    172.17.79.4      SMB2     152 Tree Connect Request, Tree: '\\DC01\Trip'
```

So to sum up The source Port the attacker used to connect to the workstation can be found in our earlier EventLog Analysis

```
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Event version         : 2
Event level           : Information (0)
Computer name         : Forela-Wkstn001.forela.local
Source name           : Microsoft-Windows-Security-Auditing
Channel name          : Security
Event identifier      : 0×00001210 (4624)
Number of strings     : 27
String: 1             : S-1-0-0
String: 2             : -
String: 3             : -
String: 4             : 0×0000000000000000
String: 5             : S-1-5-21-3239415629-1862073780-2394361899-1601
String: 6             : arthur.kyle
String: 7             : FORELA
String: 8             : 0×000000000064a799
String: 9             : 3
String: 10            : NtLmSsp
String: 11            : NTLM
String: 12            : FORELA-WKSTN002
String: 13            : {00000000-0000-0000-0000-000000000000}
String: 14            : -
String: 15            : NTLM V2
String: 16            : 128
String: 17            : 0×0000000000000000
String: 18            : -
String: 19            : 172.17.79.135
String: 20            : 40252
String: 21            : %%1833
String: 22            : -
String: 23            : -
String: 24            : -
String: 25            : %%1843
String: 26            : 0×0000000000000000
String: 27            : %%1843
```

Which is **40525** with LogonID of **0x64A799** (string 8) and the logon happens from **FORELA-WKSTN002,172.17.79.135** (string 12 & string 19) at **2024-07-31 04:55:16** as the log shows . to observe the accessed share i can filter the EventLogs for EventID 5140 which will show the accessed shares

```
┌──(kali㉿kali)-[~/Htb/Reaper/Reaper]
└─$ cat EventLogs.xml | grep '(5140)' -A 12 -B 8
```

```
Event number                    : 5
Creation time                   : Jul 31, 2024 04:55:16.243325000 UTC
Written time                    : Jul 31, 2024 04:55:16.243325000 UTC
Event version                   : 1
Event level                     : Information (0)
Computer name                   : Forela-Wkstn001.forela.local
Source name                     : Microsoft-Windows-Security-Auditing
Channel name                    : Security
Event identifier                : 0×00001414 (5140)
Number of strings               : 11
String: 1                       : S-1-5-21-3239415629-1862073780-2394361899-1601
String: 2                       : arthur.kyle
String: 3                       : FORELA
String: 4                       : 0×000000000064a799
String: 5                       : File
String: 6                       : 172.17.79.135
String: 7                       : 40252
String: 8                       : \\*\IPC$
String: 9                       :
String: 10                      : 0×00000001
String: 11                      : %%4416
```

We can see that the TimeLine fits also all other informations like the user accessed the share (the victim) it seems he accessed ***\IPC$** specifically the $IPC share . That's Close Up the Full Attack Path being Executed by The Attacker starting from LLMNR poisoning to NTLM relaying and accessing smb shares .

Remediation

To stop LLMNR poisoning and NTLM relaying, disable legacy protocols and enforce cryptographic signing across your network. Use Group Policy to turn off LLMNR and NetBIOS over TCP/IP, forcing systems to use secure DNS instead. Next, configure SMB Signing (Always) and LDAP signing to block attackers from intercepting and forwarding network hashes. Finally, enable Extended Protection for Authentication (EPA) and restrict NTLM usage to secure high-privilege accounts.